IILM UNIVERSITY

Course Pack Framework

A Comprehensive Instructional Delivery Document

OBE

(Education)

What the Student should achieve?

OBLT

(Learning and Teaching)

How to make the students to achieve the Outcome?

OBA

(Assessment)

How to measure what the students has



Prepared by:

School of Computer Science and Engineering

VISION -SCSE

To be a globally recognized institute that fosters innovation, Critical thinking approach to create sustainable technological solutions and develop ethical, future-ready responsible professionals for the evolving digital society.

MISSION – SCSE

M1. To provide quality education in Computer Science that equips students with strong theoretical foundations, practical skills, and the ability to solve real-world technological challenges using emerging technologies.

M2. To establishing state-of-the-art laboratories, research facilities and CoE that enable students and faculty to design, develop. and deploy innovative and sustainable engineering solutions.

M3. To cultivate a culture of innovation, creativity, and multidisciplinary research in emerging areas of computing to address global and societal challenges through establishment.

M4. To strengthen partnerships with industry, academic, and international collaborations to promote knowledge exchange, entrepreneurship, and impactful technological advancement.

Preamble

The Course Pack is a detailed pedagogical guide that outlines the components of instruction delivery by faculty members. It includes the course structure, course overview, objectives, prerequisite courses, program-specific outcomes (PSOs), course outcomes (COs), Bloom's Taxonomy (Knowledge Levels), types of courses, course articulation matrix, assessment patterns, course content, lesson plans, bibliography, problem-based learning or case studies, and student-centred learning (promoting self-learning and lifelong learning). The Course Pack ensures a consistent approach to course delivery across the university while allowing faculty members the flexibility to introduce innovative teaching methods and develop diverse assessment tools (including alternate assessment methods).

Each faculty member responsible for teaching a course develops the Course Pack. When multiple faculty members teach the same course, they collaborate as a team, with a senior faculty member (Course Lead) guiding the design and delivery of the course. This collaborative effort provides ample opportunities for faculty members to introduce innovative teaching strategies within the department or school.

Overall, the Course Pack serves as a comprehensive framework for teaching and learning, guiding faculty members across schools and departments in the university. It is not only a tool for evaluating student learning but also for continuously assessing the achievement of course learning outcomes for all students.

Scheme

The scheme provides an overview of work-integrated learning opportunities, offering students hands-on experience in real-world settings. It outlines what the course involves and the practical skills students will gain.

<table>
<tr><td>Course Title</td><td colspan="3">Emerging Technologies</td><td colspan="3">Course Type</td><td colspan="3">Theory</td></tr>
<tr><td>Course Code</td><td colspan="3">26SOST111</td><td colspan="3">Class</td><td colspan="3">B.Tech I /II Sem</td></tr>
<tr><td rowspan="6">Instruction Delivery</td><td>Activity</td><td>Credits</td><td>Weekly Hours</td><td colspan="3" rowspan="2">Total Number of Classes per Semester</td><td colspan="3" rowspan="2">Assessment in Weightage</td></tr>
<tr><td>Lecture</td><td>2</td><td>2</td></tr>
<tr><td>Tutorial</td><td>0</td><td>0</td><td rowspan="3"></td><td rowspan="3"></td><td rowspan="3"></td><td rowspan="3">Self-Study</td><td rowspan="3">MTE</td><td rowspan="3">SEE</td></tr>
<tr><td>Practical</td><td>0</td><td>0</td></tr>
<tr><td>Self- study</td><td>0</td><td>0</td></tr>
<tr><td>Total</td><td>2</td><td>2</td><td>45</td><td>0</td><td>0</td><td></td><td></td><td></td></tr>
<tr><td rowspan="3"></td><td>Course Lead</td><td colspan="3"></td><td colspan="2">Course Coordinator</td><td colspan="3"></td></tr>
<tr><td colspan="4">List of faculties teaching Theory</td><td colspan="5">List of faculties teaching Practical</td></tr>
<tr><td colspan="4">1.</td><td colspan="5">Not Applicable</td></tr>
</table>

Course Overview

This course introduces students to the rapidly evolving landscape of emerging technologies that are transforming the field of engineering and related industries. It provides a foundational understanding of key technological advancements such as Artificial Intelligence, Internet of Things (IoT), Cloud Computing, Cybersecurity, Data Analytics, Blockchain, Robotics, and other next-generation innovations.

The course is designed to develop interdisciplinary awareness and help students understand how these technologies interact to solve real-world engineering and societal challenges. Through conceptual discussions, case studies, and industry-oriented applications, students will gain insight into current trends, innovation ecosystems, and future career opportunities.

By the end of the course, students will be equipped with the knowledge and mindset required to adapt to technological changes, think innovatively, and prepare for the evolving demands of modern industries and engineering practices.

Pre-Requisite

This course does not require any advanced technical background. Basic knowledge of science, mathematics, and fundamental engineering concepts will be helpful. Students are expected to have an interest in technology, innovation, and emerging industry trends, along with a willingness to explore interdisciplinary applications of modern engineering technologies.

Pre-Requisite course required	**Required**	
If, yes please fill in the Details	**Course Code**	**Course Title**

Program Outcome (PO's)

Program Outcomes (POs) are typically defined by the relevant apex body or council. If a council does not prescribe specific POs, the school offering the program is responsible for designing and developing the POs in alignment with the Program Educational Objectives (PEOs). These POs must be reviewed and approved by the Board of Studies (BOS) and subsequently submitted to the Academic Council for final approval. Upon successfully completing the course, students will be able to:

PO No.	Description of the Program Outcome
PO1	**Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization to develop solutions to complex engineering problems.
PO2	**Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.
PO3	**Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.
PO4	**Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.
PO5	**Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.
PO6	**The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

PO7	**Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.
PO8	**Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO9	**Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
PO10	**Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO11	**Life-Long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

Program Specific Outcome (PSO's)

Program Specific Outcomes (PSOs) are concise statements outlining the abilities and competencies that graduates of a particular discipline-specific program are expected to achieve. Each program should define two to three PSOs.

PSO	Description of the Program Specific Outcome
PSO1	Graduates will be able to analyze, design, develop, and deploy scalable and secure software systems using modern programming languages, software engineering practices, and development frameworks to address real-world industry and societal needs.
PSO2	Graduates will be able to apply data analytics, artificial intelligence, and machine learning techniques to extract insights from complex data and develop intelligent computing solutions that support decision-making in industry and society.
PSO3	Graduates will be able to utilize emerging technologies such as cloud computing, cybersecurity, Internet of Things (IoT), data sciences and distributed systems to design innovative, ethical, and sustainable technological solutions for evolving global and societal challenges

Program Educational Objectives (PEOs)

PEO1. Graduates will apply strong foundations in computer science and emerging technologies to develop innovative and sustainable solutions to real-world problems**.**

PEO2. Graduates will pursue successful careers, higher education, professional certifications, and continuous learning to adapt to evolving technological advancements**.**

PEO3. Graduates will contribute to innovation, multidisciplinary teamwork, and industry–academia collaboration while addressing real-world and societal challenges through computing solutions**.**

PEO4. Graduates will demonstrate professional ethics, leadership, effective communication, and social responsibility, contributing to sustainable development and the global digital society.

Course Content

Course Objectives:

After the completion of the course, students will be able to

1. Explain fundamental concepts of emerging technologies.
2. Identify applications across engineering domains.
3. Analyze technological trends and interdisciplinary linkages.
4. Evaluate real-world use cases and societal impact.
5. Propose basic solutions using emerging technologies.

Course Content
Unit 1: Introduction to Emerging Technologies • Technology evolution and digital transformation • Industry 4.0 and Society 5.0 • Interdisciplinary engineering concepts
Unit 2: Cloud Computing & AI/Data Science • Basics of Cloud Computing (IaaS, PaaS, SaaS) • Introduction to Artificial Intelligence & Machine Learning • Fundamentals of Data Science & Analytics • Real-world applications
Unit 3: IoT, Robotics & Cyber-Physical Systems • Internet of Things (architecture & applications) • Basics of Robotics & Automation • Introduction to Cyber-Physical Systems (CPS) • Smart systems and Industry applications
Unit 4: Semiconductor & Advanced Computing • Basics of Semiconductor Technology • Introduction to Quantum Computing • Hardware-software integration
Unit 5: Sustainable & Future Technologies • Renewable Energy Technologies • Electric Vehicles (EV) – basics and ecosystem

- Sustainability and green engineering

Unit 6: Security & Bio-Engineering Domains

- Basics of Cyber Security
- Introduction to Biomedical Engineering
- Overview of Bioinformatics & Biotechnology Engineering
- Ethical and societal implications

Course Outcome (CO)

Course Outcome	Upon successful completion of this course, the student will be able to:
26SOST111.1	Explain fundamental concepts of emerging technologies.
26SOST111.2	Identify applications across engineering domains.
26SOST111.3	Analyze technological trends and interdisciplinary linkages.
26SOST111.4	Evaluate real-world use cases and societal impact.
26SOST111.5	Propose basic solutions using emerging technologies.

Course Articulation Matrix

The course articulation matrix represents the relationship between Course Outcomes, Program Outcomes and Program Specific Outcome, highlighting the expected strength of their alignment at three levels: low, medium, and high.

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3	PSO4
26SOST111.1	3													1	
26SOST111.2	2	3			2									3	
26SOST111.3		3			1	2					2			3	
26SOST111.4		2		3		2					2			3	
226SOST111.5	2	2	3		2						2			3	

3: High Correlation | 2: Medium Correlation | 1: Low Correlation |-: No Correlation

Taxonomy Level of the Course Outcomes

Bloom's taxonomy is a hierarchical framework used to classify educational learning objectives based on their complexity and specificity. It encompasses three primary learning domains: cognitive, affective, and psychomotor.

CO No.	Remember KL1	Understand KL 2	Apply KL 3	Analyse KL 4	Evaluate KL 2	Create KL 6
26CSEM712.1		Yes				
26CSEM712.2			Yes	Yes		
26CSEM712.3			Yes			
26CSEM712.4				Yes		
26CSEM712.5					Yes	

Evaluation Matrix

Theory

- **Continuous Internal Assessment (CIA)**
  Includes assignments and quizzes to evaluate continuous learning during the semester.
- **Mid Term Examination (MTE)**
  Mid Term examination is conducted in the middle of the semester to assess the understanding of course concepts and theoretical knowledge.
- **End Term Examination (ETE)**
  Final written examination conducted at the end of the semester to assess the overall understanding of course concepts and theoretical knowledge.

Assessment Component	Category	Marks	Weightage (%)
Attendance	Internal	5	5%
Professional Conduct/Learning Attitude	Internal	5	5%
Certification	Internal	10	10%
Viva / Presentation	Internal	10	10%
Continuous Assessment Total		30	30%
Mid-Term Test (MT/MMT)	MMT	20	20%
Total (Internal)		50	50%
End-Term Examination (ETE)	ET	50	50%
ET Total		50	50%
Grand Total		100	100%

Lecture Plan

L.No.	Project-Based Learning Activity / Topic	Skills Developed	Competency
1	Study of Emerging Technologies in Daily Life	Observation, technology awareness	Understand modern technological trends
2	Timeline Chart of Industrial Revolutions	Analytical thinking, presentation skills	Explain technology evolution
3	Case Study on Digital Transformation	Critical analysis, communication	Analyze digital transformation applications
4	Smart Factory Model Discussion	Teamwork, conceptual understanding	Understand Industry 4.0 concepts
5	Mini Survey on Automation in Industries	Research, data collection	Identify industrial automation practices
6	Society 5.0 Concept Presentation	Creativity, communication	Explain human-centered technologies
7	Comparative Study: Industry 4.0 vs Society 5.0	Comparative analysis	Analyze emerging technology frameworks
8	Interdisciplinary Engineering Case Study	Problem-solving, integration skills	Understand interdisciplinary applications
9	Seminar on Emerging Engineering Domains	Technical presentation	Improve technical communication
10	Cloud Service Provider Comparison	Research, analytical skills	Understand cloud computing services
11	Classification of Cloud Models	Concept mapping	Identify cloud service models
12	Cloud Storage Demonstration	Practical exposure	Apply cloud computing basics
13	Real-world Cloud Application Study	Case analysis	Analyze cloud applications
14	AI-based Application Identification	Technology awareness	Understand AI fundamentals
15	Simple Machine Learning Workflow Study	Logical thinking	Understand ML concepts
16	AI in Healthcare/Finance Presentation	Research and communication	Identify AI applications
17	Data Collection and Visualization Activity	Data handling	Understand data science basics
18	Big Data Application Discussion	Analytical thinking	Explain analytics applications

19	AI/Data Science Case Study Review	Critical evaluation	Evaluate real-world use cases
20	IoT Device Identification Activity	Observation, technical awareness	Understand IoT concepts
21	IoT Architecture Diagram Preparation	Design thinking	Explain IoT architecture
22	Study of IoT Communication Protocols	Technical analysis	Identify IoT communication methods
23	Smart Home/Smart Agriculture Case Study	Problem-solving	Analyze IoT applications
24	Robotics Components Demonstration	Technical understanding	Understand robotics basics
25	Industrial Automation Video Analysis	Observation and interpretation	Analyze automation systems
26	CPS Architecture Study	System analysis	Explain cyber-physical systems
27	Smart City System Discussion	Teamwork, innovation	Understand smart systems
28	Industry Automation Case Study	Analytical skills	Evaluate industrial applications
29	Semiconductor Material Identification	Technical knowledge	Understand semiconductor basics
30	Electronic Components Observation	Practical exposure	Identify semiconductor devices
31	Semiconductor Fabrication Process Study	Research skills	Explain manufacturing concepts
32	Introduction to Quantum Computing Seminar	Conceptual learning	Understand quantum computing basics
33	Qubit and Quantum Gate Activity	Analytical thinking	Explain quantum concepts
34	Quantum Computing Applications Discussion	Innovation awareness	Analyze advanced computing applications
35	Embedded System Demonstration	Technical skills	Understand hardware-software integration
36	Emerging Computing Technologies Review	Research and presentation	Identify future computing technologies
37	Renewable Energy Source Survey	Environmental awareness	Understand renewable technologies
38	Green Engineering Practices Discussion	Sustainability awareness	Analyze sustainable engineering
39	EV Components Identification	Technical understanding	Understand EV basics
40	EV Charging Infrastructure Study	Analytical and observation skills	Explain EV ecosystem
41	Smart Grid Case Study	Problem-solving	Understand smart energy systems
42	Cyber Security Threat Analysis	Risk assessment	Identify cybersecurity threats
43	Biomedical Device Demonstration	Technical awareness	Understand biomedical engineering

44	Biotechnology and Bioinformatics Seminar	Research and communication	Explain bioengineering concepts
45	Ethics and Technology Debate / Course Review	Critical thinking, teamwork	Evaluate societal and ethical implications

Reference's

Text Books:

1. "Emerging Technologies for Engineers" by Reema Thareja, 2024, ISBN **978-9363869158**
2. "Emerging Trends in Biotechnology" by Dr. Sonali Patil, Dr. Vandana Santosh Gupta, Dr. Annika Durve Gupta ISBN 978-93-90471-83-6
3. Green Technology and Sustainable Development: A Textbook for Modern Environmental Science, ISBN: 978-93-343-9459-7
4. Green Growth and Green Energy: Opportunity for Sustainable Development" by Dr. Chinmoy Sarkar, Dr. Paramita Dasgupta, et al., ISBN 978-0124114548

Reference Books:

5. Emerging Technologies in Ai, Computation, Communication, and Cybersecurity

   by Manoj Kumar , Tanweer Ali, et al.
6. J, Song I, "Emerging Technologies for Emerging Markets",14th Edition, Springer Singapore, 2014 2.
7. Sadiku M N O, "Emerging Internet-based Technologies", CRC Press, 2014
8. Del Rosal V, "Emerging Technologies and Future of Work", Emtechub, 2015

MooC Course

Coursera

1. Emerging Technologies: From Smartphones to IoT to Big Data Specialization
2. IT Infrastructure and Emerging Trends

NPTEL

1. Introduction to Industry 4.0 and Industrial Internet of Things
2. Cloud Computing
3. Fundamentals of Electric vehicles: Technology & Economics, IIT Madras

Problem-Based Learning / Case Studies / Clinics

Unit 1: Introduction to Emerging Technologies

1. Apply the concept of digital transformation to redesign a manual college admission process into a digital system.
2. Analyze the differences between Industry 3.0 and Industry 4.0 in terms of automation, connectivity, and efficiency in small industries.
3. Analyze the technological gaps between rural and urban areas and identify how emerging technologies can reduce them.
4. Evaluate the impact of Society 5.0 on healthcare, education, and transportation systems.
5. Design a conceptual model of a smart campus ecosystem integrating AI, IoT, and cloud technologies.

Unit 2: Cloud Computing, AI & Data Science

6. Apply cloud computing services (IaaS, PaaS, SaaS) to propose a storage solution for a university database system.
7. Analyze factors affecting poor student performance using basic data science concepts.
8. Analyze the working of recommendation systems used in e-learning platforms.
9. Evaluate the effectiveness of AI-based healthcare diagnosis systems compared to traditional methods.
10. Design a conceptual AI-based attendance system using face recognition or sensor data.

Unit 3: IoT, Robotics & Cyber-Physical Systems

11. Apply IoT architecture to design a smart irrigation system for agriculture fields.
12. Analyze the causes of traffic congestion and propose how IoT and CPS can improve traffic flow.
13. Analyze the role of sensors and actuators in smart home automation systems.
14. Evaluate the effectiveness of robotics in industrial automation compared to manual labour.

15. Design a smart waste management system using IoT-enabled smart bins.

Unit 4: Semiconductor & Advanced Computing

16. Apply semiconductor principles to explain the working of basic electronic devices used in smartphones.
17. Analyze limitations of classical computing in handling large-scale real-time data.
18. Analyze the architecture of embedded systems used in modern smart devices.
19. Evaluate the potential of quantum computing compared to classical computing for cryptography.
20. Design a conceptual model of an edge computing-based smart device system.

Unit 5: Sustainable & Future Technologies

21. Apply renewable energy concepts to propose a solar-based power solution for rural electrification.
22. Analyze environmental impacts of Electric Vehicles compared to fuel-based vehicles.
23. Analyze the functioning of smart grids in energy distribution systems.
24. Evaluate different renewable energy technologies for sustainable development.
25. Design a waste-to-energy conversion model for urban waste management.

Unit 6: Security & Bio-Engineering Domains

26. Apply cybersecurity principles to secure a basic online banking system.
27. Analyze common cyber threats affecting personal and organizational data systems.
28. Analyze biomedical engineering applications in modern healthcare systems.
29. Evaluate the effectiveness of biometric authentication systems over password-based systems.
30. Design a smart health monitoring system using IoT and biomedical sensors.

Unit 2: Virtualisation and Cloud Architecture

31. Compare virtual machines and containers for deploying a microservices-based application and justify the better approach.
32. Illustrate how virtualization enables resource abstraction and isolation in a multi-tenant cloud environment.
33. Design a basic container orchestration workflow using Kubernetes concepts such as pods, services, and scaling.
34. Identify challenges that may arise in a virtualized cloud infrastructure and propose suitable solutions.

Unit 3: Scheduling in Cloud Computing

35. Analyze how task scheduling algorithms such as FCFS and Round Robin affect execution time and resource utilization in cloud systems.
36. Develop a scheduling strategy for a cloud-based scientific workflow involving multiple dependent tasks.
37. Evaluate different scheduling models for handling large-scale cloud applications with multiple users.
38. Examine the role of multithreading in improving performance of cloud applications.

Unit 4: Advanced Scheduling and Optimisation

39. Propose an energy-efficient scheduling strategy for a cloud data center handling high computational workloads.
40. Analyze how QoS and SLA requirements influence scheduling decisions in enterprise cloud applications.
41. Compare cost-aware and performance-aware resource allocation strategies for cloud services.
42. Study real-world cloud platforms and evaluate how they manage resource scheduling and scalability.

Unit 5: Project-Based Implementation

43. Design and deploy a small cloud application that automatically scales resources based on workload demand.
44. Implement a workflow scheduling solution for a data analytics task using a public cloud free tier or institutional cloud environment.
45. Develop a cost-optimized deployment plan for a cloud-hosted web application.

Unit 6: Research and Emerging Trends

46. Review recent research papers on AI-driven scheduling techniques in cloud environments.
47. Investigate how edge computing and fog computing influence cloud resource management strategies.
48. Identify open research challenges in energy-efficient and SLA-aware scheduling for cloud data centers.

QUESTION BANK (300 Questions)

50 Questions from Each Unit

(50*6 = 300 Questions)

Q. No.	CO	Answer Length	Marks	BKL	Statement of the Question
1	CO1	Short	2	1	Define emerging technologies.
2	CO1	Short	2	1	What is digital transformation?
3	CO1	Short	2	1	Define Industry 4.0.
4	CO1	Short	2	1	What is Society 5.0?
5	CO1	Short	2	1	List any two characteristics of emerging technologies.
6	CO1	Short	2	1	What is technological evolution?
7	CO1	Short	2	1	State any two benefits of digital transformation.
8	CO1	Short	2	1	What is interdisciplinary engineering?
9	CO1	Short	2	1	Mention two examples of disruptive technologies.
10	CO1	Short	2	1	What role does data play in digital transformation?
11	CO1	Medium	5	3	Explain the concept of technology evolution.
12	CO1	Medium	5	3	Describe the key drivers of digital transformation.
13	CO1	Medium	5	3	Differentiate between digitization and digitalization.
14	CO1	Medium	5	3	Explain the significance of Industry 4.0.
15	CO1	Medium	5	3	Describe the objectives of Society 5.0.
16	CO1	Medium	5	3	Explain the impact of digital technologies on industries.
17	CO1	Medium	5	3	Discuss the importance of interdisciplinary engineering.
18	CO1	Medium	5	3	Explain how engineering disciplines collaborate in modern systems.
19	CO1	Medium	5	3	Discuss the role of innovation in emerging technologies.
20	CO1	Medium	5	3	Explain the concept of smart manufacturing.
21	CO1	Long	10	3	Analyze the evolution of technology from Industry 1.0 to Industry 4.0.
22	CO1	Long	10	3	Explain the stages of digital transformation with examples.
23	CO1	Long	10	3	Discuss the impact of Industry 4.0 on engineering practices.
24	CO1	Long	10	3	Compare Industry 4.0 and Society 5.0.
25	CO1	Long	10	3	Explain the technologies enabling Industry 4.0.
26	CO1	Long	10	3	Discuss interdisciplinary engineering in smart city development.
27	CO1	Long	10	3	Explain the role of automation in digital transformation.
28	CO1	Long	10	3	Analyze challenges faced during digital transformation.
29	CO1	Long	10	3	Discuss opportunities created by emerging technologies.
30	CO1	Long	10	3	Explain the role of AI, IoT, and cloud computing in digital transformation.

31	CO1	Short	2	1	What is a smart factory?
32	CO1	Short	2	1	Define cyber-physical integration.
33	CO1	Short	2	1	State two applications of Industry 4.0.
34	CO1	Short	2	1	What is innovation?
35	CO1	Short	2	1	Mention any two emerging technologies.
36	CO1	Medium	5	3	Explain the role of connectivity in Industry 4.0.
37	CO1	Medium	5	3	Discuss the importance of data-driven decision making.
38	CO1	Medium	5	3	Explain the relationship between Industry 4.0 and smart manufacturing.
39	CO1	Medium	5	3	Describe the societal benefits of Society 5.0.
40	CO1	Medium	5	3	Explain the concept of digital ecosystems.
41	CO1	Long	10	3	Evaluate the impact of digital transformation on education.
42	CO1	Long	10	3	Discuss emerging technologies shaping future industries.
43	CO1	Long	10	3	Explain how interdisciplinary engineering supports innovation.
44	CO1	Long	10	3	Analyze the economic impact of Industry 4.0.
45	CO1	Long	10	3	Discuss the role of human-centered technology in Society 5.0.
46	CO1	Long	10	3	Explain the importance of sustainability in emerging technologies.
47	CO1	Long	10	3	Analyze barriers to adopting emerging technologies.
48	CO1	Long	10	3	Discuss the future trends in digital transformation.
49	CO1	Long	10	3	Explain how emerging technologies improve productivity.
50	CO1	Long	10	3	Evaluate the future scope of interdisciplinary engineering.
51	CO2	Short	2	1	Define cloud computing.
52	CO2	Short	2	1	What is IaaS?
53	CO2	Short	2	1	Define PaaS.
54	CO2	Short	2	1	What is SaaS?
55	CO2	Short	2	1	Define Artificial Intelligence.
56	CO2	Short	2	1	What is Machine Learning?
57	CO2	Short	2	1	Define Data Science.
58	CO2	Short	2	1	What is data analytics?
59	CO2	Short	2	1	Mention two cloud service providers.
60	CO2	Short	2	1	List two applications of AI.
61	CO2	Medium	5	2	Explain the characteristics of cloud computing.
62	CO2	Medium	5	2	Differentiate between IaaS, PaaS and SaaS.
63	CO2	Medium	5	2	Explain the advantages of cloud computing.
64	CO2	Medium	5	2	Describe the basic workflow of machine learning.
65	CO2	Medium	5	2	Explain supervised learning with examples.
66	CO2	Medium	5	2	Explain unsupervised learning with examples.
67	CO2	Medium	5	2	Discuss the importance of data science.

68	CO2	Medium	5	2	Explain the role of data analytics in business.
69	CO2	Medium	5	2	Describe cloud deployment models.
70	CO2	Medium	5	2	Explain AI applications in healthcare.
71	CO2	Long	10	3	Illustrate the architecture of cloud computing.
72	CO2	Long	10	4	Analyze the benefits and challenges of cloud adoption.
73	CO2	Long	10	3	Explain the working of machine learning systems.
74	CO2	Long	10	4	Compare AI and Machine Learning.
75	CO2	Long	10	4	Analyze the role of cloud computing in digital transformation.
76	CO2	Long	10	3	Explain the stages of a data science project.
77	CO2	Long	10	4	Analyze the impact of AI on modern industries.
78	CO2	Long	10	5	Evaluate the effectiveness of cloud services in organizations.
79	CO2	Long	10	4	Discuss ethical issues in AI applications.
80	CO2	Long	10	5	Evaluate the importance of data-driven decision making.
81	CO2	Short	2	1	Define predictive analytics.
82	CO2	Short	2	1	What is a dataset?
83	CO2	Short	2	1	Define cloud storage.
84	CO2	Short	2	1	What is deep learning?
85	CO2	Short	2	1	Define neural network.
86	CO2	Medium	5	2	Explain AI in smart cities.
87	CO2	Medium	5	2	Discuss cloud security concerns.
88	CO2	Medium	5	2	Explain the significance of big data.
89	CO2	Medium	5	2	Describe AI-powered recommendation systems.
90	CO2	Medium	5	2	Explain the importance of data visualization.
91	CO2	Long	10	4	Analyze the future scope of AI and data science.
92	CO2	Long	10	5	Evaluate cloud computing for educational institutions.
93	CO2	Long	10	4	Analyze AI applications in transportation.
94	CO2	Long	10	3	Demonstrate how cloud services support startups.
95	CO2	Long	10	4	Analyze challenges in implementing AI solutions.
96	CO2	Long	10	5	Evaluate the societal impact of AI.
97	CO2	Long	10	6	Design a cloud-based solution for a small business.
98	CO2	Long	10	6	Propose a data analytics framework for a retail company.
99	CO2	Long	10	5	Assess the role of AI in future engineering applications.
100	CO2	Long	10	6	Develop a conceptual AI-based smart service application.
101	CO2	Short	2	1	Define Internet of Things (IoT).
102	CO2	Short	2	1	What is a sensor in an IoT system?
103	CO2	Short	2	1	Define actuator.
104	CO2	Short	2	1	What is robotics?
105	CO2	Short	2	1	Define automation.

106	CO2	Short	2	1	What is a Cyber-Physical System (CPS)?
107	CO2	Short	2	1	List any two applications of IoT.
108	CO2	Short	2	1	What is machine-to-machine (M2M) communication?
109	CO2	Short	2	1	Define smart system.
110	CO2	Short	2	1	Mention any two examples of industrial robots.
111	CO2	Medium	5	2	Explain the concept of IoT.
112	CO2	Medium	5	2	Describe the basic architecture of IoT.
113	CO2	Medium	5	2	Explain the role of sensors in IoT applications.
114	CO2	Medium	5	2	Discuss the functions of actuators in smart systems.
115	CO2	Medium	5	2	Explain the characteristics of IoT devices.
116	CO2	Medium	5	2	Describe the components of a robotic system.
117	CO2	Medium	5	2	Explain the importance of automation in industries.
118	CO2	Medium	5	2	Discuss the features of Cyber-Physical Systems.
119	CO2	Medium	5	2	Explain smart home applications of IoT.
120	CO2	Medium	5	2	Describe the role of IoT in healthcare.
121	CO2	Long	10	3	Illustrate the layered architecture of IoT with a neat diagram.
122	CO2	Long	10	4	Analyze the impact of IoT on industrial automation.
123	CO2	Long	10	3	Explain the working of a smart agriculture system using IoT.
124	CO2	Long	10	4	Compare traditional systems and Cyber-Physical Systems.
125	CO2	Long	10	4	Analyze the role of robotics in manufacturing industries.
126	CO2	Long	10	3	Explain the integration of IoT and cloud computing.
127	CO2	Long	10	4	Discuss the challenges in implementing IoT systems.
128	CO2	Long	10	5	Evaluate the benefits of automation in Industry 4.0.
129	CO2	Long	10	4	Analyze the significance of CPS in smart cities.
130	CO2	Long	10	5	Evaluate the role of robotics in improving productivity.
131	CO2	Short	2	1	What is an embedded system?
132	CO2	Short	2	1	Define smart factory.
133	CO2	Short	2	1	What is RFID?
134	CO2	Short	2	1	Define autonomous robot.
135	CO2	Short	2	1	What is industrial automation?
136	CO2	Medium	5	2	Explain the working principle of RFID technology.
137	CO2	Medium	5	2	Discuss the applications of autonomous robots.
138	CO2	Medium	5	2	Explain the role of embedded systems in IoT.
139	CO2	Medium	5	2	Describe the communication protocols used in IoT.
140	CO2	Medium	5	2	Explain the concept of smart manufacturing.

141	CO2	Long	10	4	Analyze security issues in IoT-based systems.
142	CO2	Long	10	5	Evaluate the effectiveness of smart transportation systems.
143	CO2	Long	10	4	Discuss the role of robotics in healthcare applications.
144	CO2	Long	10	3	Demonstrate how CPS enhances industrial operations.
145	CO2	Long	10	5	Evaluate the advantages and limitations of IoT technology.
146	CO2	Long	10	6	Design an IoT-based smart parking system.
147	CO2	Long	10	6	Propose a robotic solution for warehouse automation.
148	CO2	Long	10	6	Develop a conceptual CPS model for smart energy management.
149	CO2	Long	10	5	Assess the future scope of IoT and robotics.
150	CO2	Long	10	6	Create a smart city framework integrating IoT, robotics, and CPS technologies.
151	CO3	Short	2	1	Define semiconductor.
152	CO3	Short	2	1	What is intrinsic semiconductor?
153	CO3	Short	2	1	What is extrinsic semiconductor?
154	CO3	Short	2	1	Define doping.
155	CO3	Short	2	1	What is a diode?
156	CO3	Short	2	1	Define transistor.
157	CO3	Short	2	1	What is an Integrated Circuit (IC)?
158	CO3	Short	2	1	Mention any two semiconductor materials.
159	CO3	Short	2	1	What is Moore’s Law?
160	CO3	Short	2	1	Define quantum computing.
161	CO3	Medium	5	2	Explain the properties of semiconductors.
162	CO3	Medium	5	2	Differentiate between intrinsic and extrinsic semiconductors.
163	CO3	Medium	5	2	Explain the process of doping in semiconductors.
164	CO3	Medium	5	2	Describe the working principle of a PN junction diode.
165	CO3	Medium	5	2	Explain the operation of a transistor.
166	CO3	Medium	5	2	Discuss the importance of integrated circuits in modern electronics.
167	CO3	Medium	5	2	Explain the significance of semiconductor technology in computing.
168	CO3	Medium	5	2	Describe the evolution of semiconductor devices.
169	CO3	Medium	5	2	Explain the concept of a qubit.
170	CO3	Medium	5	2	Discuss the basic principles of quantum computing.
171	CO3	Long	10	3	Illustrate the energy band structure of conductors, semiconductors, and insulators.
172	CO3	Long	10	4	Analyze the impact of semiconductor technology on modern electronics.
173	CO3	Long	10	3	Explain the fabrication process of semiconductor devices.
174	CO3	Long	10	4	Compare classical computing and quantum computing.
175	CO3	Long	10	4	Analyze the advantages and limitations of quantum computing.

176	CO3	Long	10	3	Explain the role of semiconductors in microprocessors.
177	CO3	Long	10	4	Analyze the contribution of integrated circuits to digital systems.
178	CO3	Long	10	5	Evaluate the future prospects of semiconductor technology.
179	CO3	Long	10	4	Discuss the applications of quantum computing in various domains.
180	CO3	Long	10	5	Evaluate the significance of advanced computing technologies in Industry 4.0.
181	CO3	Short	2	1	What is hardware-software integration?
182	CO3	Short	2	1	Define microprocessor.
183	CO3	Short	2	1	What is a microcontroller?
184	CO3	Short	2	1	Define logic gate.
185	CO3	Short	2	1	What is a quantum gate?
186	CO3	Medium	5	2	Explain the role of hardware-software integration in embedded systems.
187	CO3	Medium	5	2	Differentiate between microprocessors and microcontrollers.
188	CO3	Medium	5	2	Explain the importance of logic gates in digital circuits.
189	CO3	Medium	5	2	Describe the concept of quantum superposition.
190	CO3	Medium	5	2	Explain the concept of quantum entanglement.
191	CO3	Long	10	4	Analyze the role of embedded systems in advanced computing.
192	CO3	Long	10	5	Evaluate the challenges in hardware-software integration.
193	CO3	Long	10	4	Discuss the significance of semiconductor manufacturing in economic development.
194	CO3	Long	10	3	Explain the architecture of a modern microprocessor.
195	CO3	Long	10	4	Analyze the impact of miniaturization on semiconductor devices.
196	CO3	Long	10	5	Evaluate the role of quantum computing in solving complex problems.
197	CO3	Long	10	6	Design a hardware-software integrated system for smart monitoring applications.
198	CO3	Long	10	6	Propose a conceptual quantum computing solution for optimization problems.
199	CO3	Long	10	5	Assess the future trends in semiconductor and advanced computing technologies.
200	CO3	Long	10	6	Develop a framework showing the integration of semiconductor devices, embedded systems, and advanced computing technologies.
201	CO4	Short	2	1	Define renewable energy.
202	CO4	Short	2	1	What is solar energy?
203	CO4	Short	2	1	Define wind energy.
204	CO4	Short	2	1	What is hydropower?
205	CO4	Short	2	1	Define biomass energy.
206	CO4	Short	2	1	What is geothermal energy?
207	CO4	Short	2	1	Define electric vehicle (EV).
208	CO4	Short	2	1	What is sustainability?
209	CO4	Short	2	1	Define green engineering.
210	CO4	Short	2	1	What is carbon footprint?

211	CO4	Medium	5	2	Explain the importance of renewable energy technologies.
212	CO4	Medium	5	2	Describe the working principle of solar photovoltaic systems.
213	CO4	Medium	5	2	Explain the generation of electricity using wind turbines.
214	CO4	Medium	5	2	Discuss the advantages of hydropower generation.
215	CO4	Medium	5	2	Explain the role of biomass in sustainable energy production.
216	CO4	Medium	5	2	Describe the key components of an electric vehicle.
217	CO4	Medium	5	2	Explain the concept of EV charging infrastructure.
218	CO4	Medium	5	2	Discuss the importance of sustainability in engineering practices.
219	CO4	Medium	5	2	Explain the principles of green engineering.
220	CO4	Medium	5	2	Describe the environmental benefits of renewable energy.
221	CO4	Long	10	3	Illustrate the working of a solar power generation system.
222	CO4	Long	10	4	Analyze the advantages and limitations of wind energy.
223	CO4	Long	10	3	Explain the role of renewable energy in reducing carbon emissions.
224	CO4	Long	10	4	Compare renewable and non-renewable energy sources.
225	CO4	Long	10	4	Analyze the impact of renewable energy on sustainable development.
226	CO4	Long	10	3	Explain the ecosystem of electric vehicles.
227	CO4	Long	10	4	Analyze the challenges in the adoption of electric vehicles.
228	CO4	Long	10	5	Evaluate the effectiveness of government policies promoting EVs.
229	CO4	Long	10	4	Discuss the role of battery technology in EV performance.
230	CO4	Long	10	5	Evaluate the contribution of renewable energy to energy security.
231	CO4	Short	2	1	What is a smart grid?
232	CO4	Short	2	1	Define energy efficiency.
233	CO4	Short	2	1	What is a lithium-ion battery?
234	CO4	Short	2	1	Define green technology.
235	CO4	Short	2	1	What is sustainable development?
236	CO4	Medium	5	2	Explain the concept of smart grids in energy management.
237	CO4	Medium	5	2	Discuss the significance of energy storage systems.
238	CO4	Medium	5	2	Explain the role of lithium-ion batteries in EVs.
239	CO4	Medium	5	2	Describe sustainable practices in engineering projects.
240	CO4	Medium	5	2	Explain the relationship between sustainability and green engineering.
241	CO4	Long	10	4	Analyze the role of smart grids in future energy systems.
242	CO4	Long	10	5	Evaluate the environmental impact of electric vehicles.
243	CO4	Long	10	4	Discuss the importance of energy conservation in sustainable development.

244	CO4	Long	10	3	Explain methods for improving energy efficiency in industries.
245	CO4	Long	10	5	Evaluate the feasibility of renewable energy adoption in developing countries.
246	CO4	Long	10	6	Design a renewable energy solution for a rural community.
247	CO4	Long	10	6	Propose an EV charging network for a smart city.
248	CO4	Long	10	6	Develop a green engineering strategy for sustainable industrial growth.
249	CO4	Long	10	5	Assess the future scope of sustainable and renewable technologies.
250	CO4	Long	10	6	Create a framework integrating renewable energy, EVs, and sustainability for a smart city ecosystem.
251	CO5	Short	2	1	Define Cyber Security.
252	CO5	Short	2	1	What is malware?
253	CO5	Short	2	1	Define phishing.
254	CO5	Short	2	1	What is a firewall?
255	CO5	Short	2	1	Define encryption.
256	CO5	Short	2	1	What is Biomedical Engineering?
257	CO5	Short	2	1	Define Bioinformatics.
258	CO5	Short	2	1	What is Biotechnology Engineering?
259	CO5	Short	2	1	Define DNA sequencing.
260	CO5	Short	2	1	What are ethical implications of technology?
261	CO5	Medium	5	2	Explain the importance of Cyber Security in the digital age.
262	CO5	Medium	5	2	Describe different types of cyber threats.
263	CO5	Medium	5	2	Explain the role of firewalls in network security.
264	CO5	Medium	5	2	Discuss the importance of encryption in data protection.
265	CO5	Medium	5	2	Explain common cyber security practices for organizations.
266	CO5	Medium	5	2	Describe the scope of Biomedical Engineering.
267	CO5	Medium	5	2	Explain the applications of biomedical instruments.
268	CO5	Medium	5	2	Discuss the role of Bioinformatics in biological research.
269	CO5	Medium	5	2	Explain the significance of Biotechnology Engineering in healthcare.
270	CO5	Medium	5	2	Discuss ethical concerns associated with emerging technologies.
271	CO5	Long	10	3	Explain the architecture of a secure cyber security framework.
272	CO5	Long	10	4	Analyze the impact of cyber attacks on organizations.
273	CO5	Long	10	3	Explain methods used for protecting digital information systems.
274	CO5	Long	10	4	Compare different cyber security threats and mitigation techniques.
275	CO5	Long	10	4	Analyze the challenges faced in maintaining cyber security.
276	CO5	Long	10	3	Explain the working principles of biomedical diagnostic equipment.

277	CO5	Long	10	4	Analyze the role of biomedical engineering in modern healthcare.
278	CO5	Long	10	5	Evaluate the contribution of bioinformatics to genomics research.
279	CO5	Long	10	4	Analyze the applications of biotechnology in agriculture and medicine.
280	CO5	Long	10	5	Evaluate ethical issues arising from biotechnology advancements.
281	CO5	Short	2	1	What is a cyber attack?
282	CO5	Short	2	1	Define antivirus software.
283	CO5	Short	2	1	What is a biomedical sensor?
284	CO5	Short	2	1	Define genome.
285	CO5	Short	2	1	What is genetic engineering?
286	CO5	Medium	5	2	Explain the role of antivirus software in cyber security.
287	CO5	Medium	5	2	Discuss various authentication techniques used in cyber security.
288	CO5	Medium	5	2	Explain the working of biomedical sensors.
289	CO5	Medium	5	2	Describe the applications of genetic engineering.
290	CO5	Medium	5	2	Explain the societal impact of biotechnology innovations.
291	CO5	Long	10	4	Analyze the importance of cyber security awareness among users.
292	CO5	Long	10	5	Evaluate strategies for preventing cyber attacks in organizations.
293	CO5	Long	10	4	Discuss the role of artificial intelligence in cyber security.
294	CO5	Long	10	3	Explain the integration of biomedical devices with healthcare systems.
295	CO5	Long	10	4	Analyze the impact of bioinformatics on personalized medicine.
296	CO5	Long	10	5	Evaluate the benefits and risks of genetic engineering technologies.
297	CO5	Long	10	6	Design a cyber security policy for a healthcare organization.
298	CO5	Long	10	6	Propose a biomedical engineering solution for remote patient monitoring.
299	CO5	Long	10	6	Develop a bioinformatics-based framework for disease prediction.
300	CO5	Long	10	6	Create a comprehensive model addressing cyber security, biomedical engineering, and ethical considerations in future healthcare systems.

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SUSTAINABLE DEVELOPMENT GOALS